

Fund the engine swap now. Hold the platform bet.

DECISION A • FIRM • ACT THIS MONTH

Replace Bravo's unsupported workflow engine and lift Spring Boot in one piece of work. 18–33 engineer-days. No licence.

DECISION B • OPEN • GATED ON EVIDENCE

Do not place the next product family yet. Three measurements decide it. None has been taken.

The brief in four lines

[PROBLEM] Bravo runs on two dead runtimes, and its engine has already executed foreign code. Camunda 7 CE lost support on 14 Oct 2025; Spring Boot 3.5.x OSS on 30 Jun 2026. 61 injected process definitions are still loadable in production. One of them ran.	[ARCHITECTURAL TRADE-OFF] The modelling paradigm is not the deciding variable. Engineering discipline is. BPMN against GSM shaped roughly 6% of Bravo and 10% of LORA. The other ~90% looks the same on both. Bravo fails safely but invisibly; LORA fails visibly but never gives up.
[COST & VISIBILITY IMPACT] Bravo is far cheaper and carries the money. It is also blind. 5–6.5× cheaper per application and ≈21× the business value — against zero traces, no loan id in logs and a frozen end-to-end suite. The fix is about 7 engineer-days, not a re-platform.	[RECOMMENDED DECISION] Take A unconditionally. Defer B until two cheap measurements land. A costs 18–33 engineer-days and buys a supported runtime. B is genuinely open: the evidence splits, and the two measurements that would close it cost days.

Sequence. Close the RCE exposure **this week** → fund the engine and Spring Boot upgrade **this month** → take gating measurements 1 and 2, which cost **days** → place the next product family. Only the last step is deferred, and only on evidence.

A does not wait for B. B does not block A.

Bravo originates the retail book for at least another 12 months on every path in this deck. A Temporal rewrite is 12–18 months; a LORA migration 15–24. The Spring Boot upgrade is owed in all three.

Two origination platforms in production. Neither decision retires either one.

BRAVO — THE INCUMBENT — Stack. Java 17, Spring Boot 3.5.16, an embedded Camunda 7.23 CE engine and PostgreSQL, all inside one service — <code>ms-bpm</code> — behind three operator consoles. — Model. The process is <i>drawn</i>, in BPMN, and executed by the engine. 53 BPMN files. — Load. 118,253 applications in August — 41% of volume, roughly 95% of business value. Four families live, a fifth in UAT.	LORA — THE REPLACEMENT — Stack. Go services on Temporal Cloud, spread over 13 repositories. — Model. Nothing is drawn. A custom planner (GSM) derives the next runnable step from the data a loan already holds. — Load. 171,479 applications in August — 59% of volume, ≈Rp 2.4bn of value. One family live out of seven built.
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Migration state. The two run side by side and have since July 2026. **Every path in this deck leaves Bravo originating the retail book for at least another 12 months.** So neither decision is about finishing the migration. A keeps Bravo safe to operate. B places the *next* product family.

TERM	MEANS	TERM	MEANS
<code>BPMN</code>	The flowchart standard Bravo’s process is drawn in	<code>GSM</code>	LORA’s alternative. Nothing is drawn; the data decides what runs next
<code>Camunda 7</code>	Bravo’s embedded engine. We run the free Community Edition , now out of support	<code>Temporal</code>	LORA’s durable-execution engine, bought as a cloud service
<code>LOS</code>	Loan Origination System — application through to go-live	<code>ms-bpm</code>	The single Bravo service holding the engine and every human-task code path

Two runtimes past end of support. An engine that has already run someone else’s code.

COMPONENT	BRAVO RUNS	SUPPORT STATUS
Camunda 7 Community Edition	7.23.0	Ended 14 Oct 2025. The line is closed and the repository archived. There will never be another security patch.
Spring Boot 3.5.x (OSS)	3.5.16	Ended 30 Jun 2026. Commercial extension exists; we hold none.
Java 17 (Oracle premier)	17	Ends 30 Sep 2026. Unchanged by either fork.
Camunda 7.23 Enterprise	no licence held	Would end 13 Oct 2026 regardless. Not an escape route.

Verified in the code: `pom.xml:26` pins 7.23.0. No Camunda BOM, no enterprise artifact, no licence key.

61 INJECTED PROCESS DEFINITIONS REMAIN ACTIVE IN PRODUCTION

Textbook remote-code-execution attempts, uploaded across 7 sessions between 2026-05-23 and 2026-06-30. One executed and read back the hostname it ran on. All 61 are still startable.

The deploy path `/engine-rest/**` does require a credential, so this was a shared or leaked secret, not an open door. That secret — `INTERNAL_SERVICE_KEY` — grants full engine control. Separately, `SecurityConfig.java:65` leaves `/camunda/**` on `permitAll()`.

An unpatchable engine that still accepts uploads is an operational risk, not an architecture question.

This is why Decision A is carved out and sequenced first. Nothing in the platform debate changes it, and nothing in the platform debate should delay it.

What Decision B decides — and what it does not

B places the **next product family** for a 5–10 year horizon. Nothing else. Several arguments that circulate under B are already settled and should not be re-opened in the room.

IN SCOPE — GENUINELY OPEN	OUT OF SCOPE — ALREADY SETTLED
— Operability. Bravo’s workflow layer emits no tracing. Of 38 monitors, none watch a loan’s progress. — Testability. Bravo’s one scheduled end-to-end run cannot report a failure. Its suite has not changed since November 2023. — Compounding. Every family added to Bravo enlarges both gaps. — Hiring. Java, Spring and Camunda are commodity skills. Go plus an in-house planner is not. This is the one constraint headcount cannot move.	— BPMN against GSM. Assessed on slide 06. Real, and not decisive. — Does Bravo work. Yes. 118,253 applications in August, with bounded failure and no zombie class. — Unit cost. Bravo, by 5–6.5× per application. Settled. — Accelerating Bravo’s retirement. Not recommended, on any reading of the evidence. — Team sizes. A lever you set, not a property of either platform. — The engine upgrade. That is A, and it already has an answer.

Judge a modelling paradigm by what teams ship under deadline pressure, not by what the spec permits.

We accept the Bravo team’s review: the flowchart sprawl comes from how Bravo was modelled, and standard remedies exist. Of the three remedies, one is 5.5% **rolled out**, one is **not started**, and one is **not enforced**.

Where each platform actually wins

BRAVO

- **5–6.5× cheaper per application**
Rp 490–515 against Rp 2,500–3,200. The workflow tier costs Rp 58M a month against Rp 431M.
- **≈21× the business value**
Rp 50.2bn against Rp 2.4bn — about 30× more per application.
- **Four live families, not one**
Plus DF2W in UAT. Six of LORA’s seven have never taken an application.
- **A change touches one or two repositories**
LORA needs five, in strict order: schema, handler, activity, permissions, front end.
- **Commodity hiring profile**
Java, Spring, Camunda. LORA needs Go plus a planner nobody can be hired for.
- **Fleet-wide questions are answerable**
“Which loans are stuck in survey?” is one SQL query. LORA cannot answer it at all.
- **Failures are bounded and classified**
No zombie class. Failed loans park in an operator console.
- **Upstream faults are attributable**
Bravo keeps 25 upstreams separate. LORA puts ~300 behind one route.

LORA

- **3.1× fewer manual interventions**
0.126% of applications against 0.389% — 1 in 790 against 1 in 257. One ticket type accounts for the gap; it was fixed on 2026-09-10.
- **Every step traced, carrying the loan id**
Bravo’s workflow layer emits no tracing whatsoever.
- **A failure identifies a customer**
The workflow id is the application. Bravo’s logs carry no application id.
- **Marginal volume is near-free**
Volume rose 23% for 3% more spend. Bravo’s unit cost rises as volume drains away.
- **Deploys do not disturb running loans**
Each version has its own queue. On Bravo a shared sub-process redeploy reaches 73% of in-flight volume.
- **An end-to-end test that can fail**
Mostly red overnight — which at least proves it runs. Bravo’s cannot fail and is frozen at 2023.
- **Upstream failure is injectable in test**
Two lines of code. Bravo has never exercised its 50 error handlers or 190 escalations.
- **Product isolation by construction**
Separate documents, separate queues. Proven on one family only.

Deliberately excluded. Team size — 25 against 22 — is your lever, not a platform fact. Ticket volume flipped once Bravo’s full scope was counted. Cycle time is 2 days on both. Neither side enforces coverage. Full 40-row table in `compare.md` §1.

BPMN against GSM is assessed. It is not the deciding variable.

VERDICT ON LORA'S ORIGINAL DESIGN RATIONALE

The modelling paradigm shaped about **6%** of Bravo and **10%** of LORA — the automated pipeline, where **LORA** is genuinely better. The remaining **~90%** is near-identical on both. What differs across that 90% is **engineering discipline**, not paradigm.

THE RATIONALE CLAIMED	WHAT THE CODE SHOWED	VERDICT
Flowcharts sprawl with every edge case	True here. <code>ndf4w.bpmn</code> carries 111 service tasks, 137 gateways, 9,494 lines.	● LORA
Loan events are concurrent; flowcharts model that badly	Bravo is sequential by choice. Camunda supports what is missing. Nobody drew it.	● LORA
Workflow data scatters across process variables	Not how Bravo built it. Business data sits in 238 JPA entities . What scatters is lifecycle state.	● REFUTED
Temporal retries natively; engines need hand-built plumbing	Bravo built the plumbing and landed a better policy : bounded retries, a degraded last attempt, an operator queue.	● BRAVO
Compensation belongs in the execution layer	Bravo has no compensation events. LORA rolls back document fields only. Both fall back on sweepers and people.	● NEITHER

SEVEN WEAKNESSES ARE IDENTICAL ON BOTH PLATFORMS

Human-task complexity, untested orchestration, an abandoned end-to-end suite, an unenforced lifecycle, absent compensation logic, the same misleading health model, and surveyor assignment as the top operational complaint. **A platform move carries all seven across.** Full write-up in `compare-architecture.md`.

What each platform does when things break

This is the trade that matters for a 5–10 year horizon: not which paradigm is more elegant, but which failure mode your operations team can live with.

FAILURE MODE	BRAVO	LORA	SAFER
Retry policy	Bounded retries, a degraded final attempt, then an operator queue. No zombie class.	Uncapped retry. About half of loans never terminate , while every health signal stays green.	● BRAVO
Undo a failed step	No compensation events at all.	Rolls back document fields only.	● NEITHER
Deploy blast radius	Redeploying a shared sub-process reaches loans already in flight – 73% of volume , with no migration plan.	Each version owns its queue. Running loans are untouched.	● LORA
Upstream attribution	25 upstreams kept separate, so a degrading dependency is named.	~300 behind one route. You cannot tell which is failing until a merged change ships.	● BRAVO
Finding the customer	Workflow logs carry no application id . A failure is a stack trace, not a loan.	The workflow id <i>is</i> the application.	● LORA
Fleet-wide query	“Which loans are stuck in survey?” – one SQL query across <code>ACT_</code> tables.	Not answerable. State lives per workflow.	● BRAVO

Bravo fails safely but invisibly. LORA fails visibly but never gives up.

Both are fixable and both have been scoped. Bravo’s blindness is a funding decision. LORA’s non-termination is a retry-policy defect. **Neither is a property of the modelling paradigm.**

LORA carries the volume. Bravo carries the money.

Every rate here is per application, from the August billing sheet. On volume LORA holds 59% to Bravo's 41%. **On business value it inverts.**

APPLICATIONS, AUG 2026



BUSINESS VALUE



VALUE PER APPLICATION

Bravo ≈ Rp 425,000

LORA ≈ Rp 14,000

≈30× more value per application on Bravo, and ≈21× in total.

ALL OF LORA'S 59% IS A SINGLE PRODUCT FAMILY

Only NDF is in production — NDF2W 90.2%, NDF4W 9.8%. **Six of seven families are not deployed.** Three have no `deploy-prod.yaml` at all. There is no switch to flip.

BRAVO HAS FOUR FAMILIES LIVE AND A FIFTH IN UAT

NDF2W, NDF4W, RO and DF4W in production. **DF2W is in UAT** with a live LOS pen test and open go-live epics. Families never launched prove nothing about either platform — which cuts both ways.

BOTH RUPIAH FIGURES ARE A CTO-OFFICE ESTIMATE

Neither platform holds agreement, disbursement, new-money or revenue data. Extracting that split from a system of record, with period and basis written down, is **gating measurement 1**.

Bravo's workflow layer is unobservable. That is an unfunded decision, not a platform law.

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TRACES EMITTED BY BRAVO'S WORKFLOW LAYER

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OF 38 MONITORS THAT WATCH A LOAN'S PROGRESS

2023

LAST CHANGE TO THE END-TO-END SUITE. THE SCHEDULED RUN CANNOT FAIL

~7 days

ENGINEERING TO CLOSE THE HIGHEST-YIELD HALF OF THE GAP

WHAT IS MISSING, PRECISELY

No distributed tracing from the job executor. **No application id in workflow logs**, so a failure cannot be tied to a customer without manual correlation. No loan-progress monitors. The scheduled end-to-end run is wired so a failure cannot surface, and its 349 feature files are frozen at November 2023.

Operationally this means failures reach you as **tickets from branches**, not as alerts.

WHAT CLOSING IT COSTS

The two test layers that catch the most — L0 and L1 — run in seconds and are **about 7 engineer-days between them**. The 349 existing feature files already cover the journeys; they need unfreezing, not rewriting. Add job-executor instrumentation and 4xx health gates and the observability half of the platform argument disappears.

Weeks, not quarters. And far cheaper than moving products to buy visibility.

THE LARGER COST LINE IS LOGGING, NOT WORKFLOW

Logging runs at **Rp 636M a month across three platforms** — Cloud Logging Rp 271.8M, Coralogix Rp 253.3M, Datadog Rp 110.6M. That is about **11× the entire Rp 58M tier** a Bravo migration would ever retire.

Rp 81–105M a month is recoverable from platform configuration alone — non-prod retention and exclusion filters, Cloud SQL audit logs off in non-prod, flow-log sampling. No code change, no squad sprint.

Visibility is the strongest argument against Bravo, and the cheapest to neutralise.

Configuration money comes before migration money. The logging saving alone exceeds everything retiring Bravo's workflow tier would deliver.

Decision A: swap the engine. 18–33 engineer-days, no licence.

Two Apache-2.0 forks of Camunda 7 are viable. Both run on **Spring Boot 4**, both keep the existing `ACT_` schema, and both accept all 53 current BPMN files and the legacy `camunda:` namespace. Java 17 is unaffected. For comparison, a Temporal rewrite or a LORA migration is **31–57 engineer-months**.

	OPERATON 2.1.4	CIB SEVEN 2.2.0
Commits, last 12 months	2,310	258
Distinct contributors	52	31
Bus factor – share by one author	~80%	25%
Ex-Camunda engineers on board	none	two
Commercial support purchasable	no – 6-month release support	yes, terms unpublished
Interim hop to 7.24 required	yes	no

THE ONLY OPEN QUESTION INSIDE A

Do we require a support contract we can actually buy?

Yes → CIB seven. No → Operaton.

OPEN THE CIB CONVERSATION BEFORE CHOOSING, NOT AFTER

CIB publishes no contract length, service levels or price. That has to be asked directly, and it is the input the fork choice turns on.

FOUR DATABASE PREFLIGHT CHECKS GATE THE CUTOVER DATE

`ACT_GE_SCHEMA_LOG` may still hold a pre-7.22 version label. If it does, **the new engine will not start**. Hours of work. Do it before committing to a date.

Decision B: three measurements gate it. None has been taken.

#	MEASURE	IF IT LANDS ONE WAY	IF IT LANDS THE OTHER	COST
1	Agreement, new-money and disbursement split by platform – from a system of record, with period and basis written down.	Confirms the ≈21× value gap. B tilts to Bravo for anything revenue-bearing, and LORA’s volume argument disappears.	Conversion is comparable. B tilts back to LORA.	days
2	Confirm the Surveyor Platform - Release reject fix – deployed 2026-09-10 by squad LN .	It was a defect. Bravo’s rate falls to ≈0.134% against LORA’s 0.126%, and LORA’s reliability lead disappears.	It is intrinsic to how Bravo operates. The 3.1× gap stands.	hours
3	Cost to launch one un-deployed LORA family – three of six have no deploy-prod.yaml .	Cheap. “Put the next family on LORA” becomes a priced proposal rather than an assumption.	Expensive. Nobody has launched a family on LORA since NDF.	weeks
Measurements 1 and 2 are the gate. Measurement 3 prices the proposal but does not block the decision.				

THE ONE ITEM HERE THAT MUST NOT WAIT FOR EVIDENCE

Fund Decision A now. Once the engine and Spring Boot are current, B can be argued on architecture and economics alone rather than under an operational deadline. 18–33 engineer-days.

Whichever way B goes, you are buying a known liability

LEVER	NEXT FAMILY ON BRAVO	NEXT FAMILY ON LORA
DEPENDENCY	A community fork. Operaton supports each release 6 months ; CIB sells a contract but will not publish terms.	A paid Temporal Cloud contract, already in place and already funded.
OBSERVABILITY	The new family is as blind as the current four until traces and monitors are funded. Failures arrive as tickets.	Every step traced with the loan id from day one. Failures arrive as alerts.
DEPLOY SAFETY	A shared sub-process redeploy reaches 73% of in-flight volume , with no migration plan.	Versioned queues. Running loans are untouched by a deploy.
CHANGE COST	One or two repositories per change — the lowest cost per change in the estate .	Five repositories, in order , for every integration change. A generator would fix it; it is designed, not built.
PRODUCT ISOLATION	Five separate mechanisms distinguish products. Adding one touches all five, and no test shows which products a change hits.	Separate documents and queues by construction — proven on one family only .
PROOF AT SCALE	Four live families, 118,253 applications in August, at Rp 490–515 each.	One live family . Six never deployed, three with no <code>deploy-prod.yaml</code> . Rp 2,500–3,200 each.

LORA’s risk is that it is unproven at scale. Bravo’s risk is discipline that has not been funded.

Both are bounded, both are already scoped, and neither is a reason to move on its own. That is why B is still open — and why it should stay open for days, not quarters.

Take A now. Defer B. Approve four things.

DECISION A · TAKE IT NOW

Fund the engine and Spring Boot upgrade. Close the RCE exposure this week.

18–33 engineer-days. No licence. Owed on every path.

DECISION B · DEFER

Place no new family until measurements 1 and 2 are in.

They cost **days**. Either one could settle it outright.

1	Close the Camunda exposure this week. Rotate <code>INTERNAL_SERVICE_KEY</code> . Enable engine authentication and remove <code>permitAll()</code> from <code>/camunda/**</code> . Delete all 61 definitions. Identify the uploader from deploy logs, 2026-05-23 to 2026-06-30.	THIS WEEK EVERY OPTION
2	Fund the engine and Spring Boot upgrade. Answer one question first — do we require purchasable support? Then run the four <code>ACT_</code> preflight checks before committing a cutover date.	18–33 ENG-DAYS RP 45–125M
3	Commission the agreement and new-money split by platform, with period and basis recorded, and a written definition of what counts as a Bravo or a LORA application. The $\approx 21\times$ value ratio is still only a CTO-office estimate.	DAYS GATES B
4	Approve the logging configuration programme. Rp 81–105M a month from platform configuration, available in about two weeks, with no code change. That exceeds everything retiring Bravo’s workflow tier would save. Independent of B. <i>(Revised 2026-09-13: previously stated as Rp 50–90M from a Feign change; that mechanism is not confirmed active in production.)</i>	2 WEEKS RP 81–105M / MO

ONE STANDING INSTRUCTION

Bravo carries most of the book and most of the money. Resource it as a strategic platform, not as a system being wound down. Every correction this analysis made moved in its favour.

Where the numbers come from, and what is still weak

MEASURED FOR THIS ANALYSIS

Jira — projects `DF`, `D2W`, `BL`, `BLCS`, `LN`.

Datadog — tracing, 1.1M log lines a week, the full monitor list, Synthetics, CI Visibility, RUM from four consoles.

GCP billing via the FinOps API, all of August 2026.

Both codebases — `bravo-bpm-service` at `v2.93.43`, three Bravo consoles, `bravo-e2e-test`, 13 LORA repositories.

OTRS ticket export for both platforms over the same window.

KNOWN WEAKNESSES

Application counts come from a billing sheet nothing else corroborates. In every per-application figure the numerator is verified and the denominator is disputed. Ask 3 fixes this.

Survivorship. As work moves off Bravo, what remains is the difficult residue. Not corrected for.

One ticket type carries the reliability finding. Six named endpoints, squad `LN` — hours of instrumentation, not an open-ended search.

Rp 50.2bn and Rp 2.4bn are a CTO-office estimate. Period and basis unconfirmed.

READ FURTHER

`compare.md` — the 40-row comparison, sourced per row, with the recommendation in full.

`compare-architecture.md` — BPMN against GSM in detail, plus the Bravo team's review.

`option-summary.md` — Decision A in full, including the fork comparison.

Five findings packs — people, testing, observability, cost, delivery. Each has its own deck.

`ticket-analysis.md` — intervention rates on both platforms.

A has a firm answer at 18–33 engineer-days. B is open, and the two measurements that would settle it cost days.

Four earlier readings in this pack turned out to be wrong, and are corrected here. Bravo's end-to-end suite does exist, but is frozen. Team size was used as evidence, which is circular. "Seven LORA families" was really one. Bravo's consoles and the `LN` project were never counted. **Every one of those corrections moved in Bravo's favour.** The findings that survived — the dead engine, the blind workflow layer, the cost ratio and the intervention gap — are the ones to trust.